



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2008
TRIAL
HIGHER SCHOOL CERTIFICATE

Mathematics Extension 1

General Instructions

- Reading Time – 5 Minutes
- Working time – 2 Hours
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators may be used.
- Start each question in a new booklet
- The questions are of equal value
- Marks may NOT be awarded for messy or badly arranged work.
- All necessary work should be shown in every question.
- Full marks will NOT be given unless the method of the solution is shown.

Total Marks – 84

- Attempt all questions

Examiner: *R. Boros*

This is an assessment task only and does not necessarily reflect the content or format of the Higher School Certificate

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

Start each question in a new answer booklet.

Question 1 (12 marks).

Marks

- a) Find the acute angle between the intersection of the curves $y = x^2 + 4$ and $y = x^2 - 2x$, correct to the nearest minute. 2
- b) A is the point $(-4, 2)$ and B is the point $(3, -1)$. Find the coordinates of the point P which divides the interval AB externally in the ratio $2:1$ 2
- c) Differentiate $y = \log_e (\sin^{-1} x)$ 2
- d) Solve the inequality $\frac{x-1}{x+3} \geq -2$ 2
- e) If $\cos A = \frac{7}{9}$ and $\sin B = \frac{1}{3}$ where A and B are acute angles,
Prove that $A = 2B$. 2
- f) Use the substitution $u = t + 1$ to evaluate $\int_0^1 \frac{t}{\sqrt{t+1}} dt$ 2

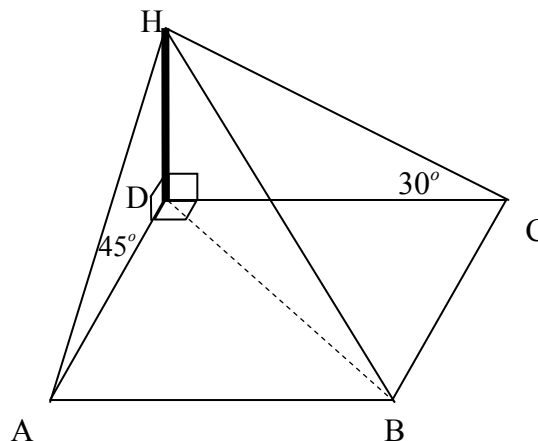
End of Question 1.

Start a new booklet.

Question 2 (12 Marks).

Marks

- a) The polynomial $P(x) = ax^3 + bx^2 - 8x + 3$ has a factor of $(x-1)$ and leaves a remainder of 15 when divided by $(x+2)$. Find the values of a and b and hence fully factorise $P(x)$. 3
- b) (i) Express $3\sin\theta + 2\cos\theta$ in the form $R\sin(\theta + \alpha)$ where α is an acute angle.
(ii) Hence, or otherwise solve the equation $3\sin\theta + 2\cos\theta = 2.5$ for $0^\circ \leq \theta \leq 360^\circ$. Answer correct to the nearest minute. 4
- c) A post HD stands vertically at one corner of a rectangular field $ABCD$. The angle of elevation of the top of the post H from the nearest corners A and C are 45° and 30° respectively.



- (i) If $AD = a$ units, find the length of BD in terms of a 2
- (ii) Hence, find the angle of elevation of H from the corner B to the nearest minute. 1
- d) Taking $x = \frac{-\pi}{6}$ as a first approximation to the root of the equation $2x + \cos x = 0$, use Newton's method once to show that a second approximation to the root of the equation is $\frac{-\pi - 6\sqrt{3}}{30}$. 2

End of Question 2.

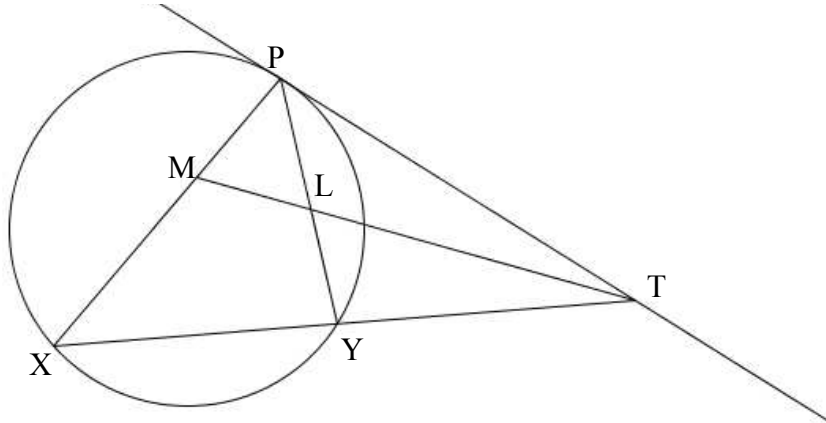
Start a new booklet.

Question 3 (12 marks).

Marks

a)

Diagram not to scale.



XY is any chord of a circle. XY is produced to T and TP is a tangent to the circle. The bisector of $\angle PTX$ meets XP in M and cuts PY at L . Prove that $\triangle MPL$ is isosceles.

3

b)

(i) Find the domain and range of $f^{-1}(x) = \sin^{-1}(3x - 1)$.

2

(ii) Sketch the graph of $y = f^{-1}(x)$.

1

(iii) Find the equation representing the inverse function $f(x)$ and state the domain and range.

3

c) Newton's Law of Cooling states that the rate of cooling of a body is proportional to the excess of the temperature of a body above the surrounding temperature. This rate can be represented by the differential equation

$$\frac{dT}{dt} = -k(T - T_0), \text{ where } T \text{ is the temperature of the body, } T_0 \text{ is the}$$

temperature of the surroundings, t is the time in minutes and k is a constant.

(i) Show that $T = T_0 + Ae^{-kt}$, where A is a constant, is a solution to the differential equation $\frac{dT}{dt} = -k(T - T_0)$.

1

(ii) A cup of coffee cools from 85°C to 80°C in one minute in a room temperature of 25°C . Find the temperature of the cup of coffee after a further 4 minutes have elapsed. Answer to the nearest degree.

2

End of Question 3.

Start a new booklet.

Question 4 (12 marks).**Marks**

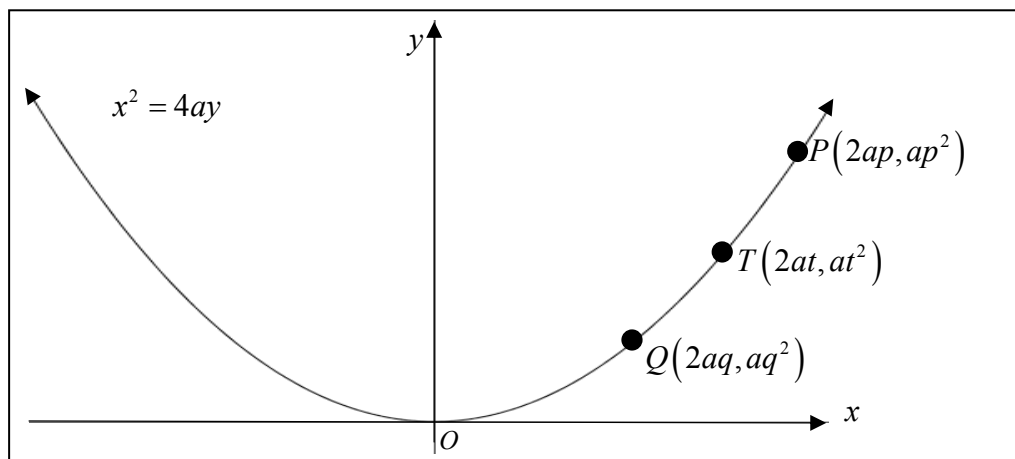
a) Find the number of ways of seating 5 boys and 5 girls at a round table if:

- (i) A particular girl wishes to sit between two particular boys.
- (ii) Two particular persons do not wish to sit together.

1

1

b) $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are the points on the parabola $x^2 = 4ay$



It is given that the chord PQ has the equation $y - \frac{1}{2}(p+q)x + apq = 0$

- (i) Derive the equation of the tangent to the parabola $x^2 = 4ay$ at the point $T(2at, at^2)$.
- (ii) The tangent at T cuts the y -axis at the point R . Find the coordinates of the point R .
- (iii) If the chord PQ passes through the point R show that p , t and q are terms of a geometric series.

2

1

2

c) A particle moves so that its distance x cm from a fixed point O at time t seconds is $x = 2 \cos 3t$.

- (i) Show that the particle satisfies the equation of motion $\ddot{x} = -n^2x$ where n is a constant.
- (ii) What is the period of the motion?
- (iii) What is the velocity when the particle is first 1 cm from O .

2

1

2

End of Question 4.

Start a new booklet.

Question 5 (12 marks).		Marks
a) Find the general solution of the equation $\tan \theta = \sin 2\theta$		3
b) The cubic equation $2x^3 - x^2 + x - 1 = 0$ has roots α, β and γ . Evaluate		
(i) $\alpha\beta + \beta\gamma + \alpha\gamma$		1
(ii) $\alpha\beta\gamma$		1
The equation $2\cos^3 \theta - \cos^2 \theta + \cos \theta - 1 = 0$ has roots $\cos a, \cos b$ and $\cos c$. Using appropriate information from parts (i) and (ii), prove that $\sec a + \sec b + \sec c = 1$.		2
c)		
(i) Sketch the curve $y = 2\cos x - 1$ for $-\pi \leq x \leq \pi$. Mark clearly where the graph crosses each axis.		2
(ii) Find the volume generated by the rotation through a complete revolution about the x axis of the region between the x -axis and that part of the curve $y = 2\cos x - 1$ for which $ x \leq \pi$ and $y \geq 0$		3

End of Question 5

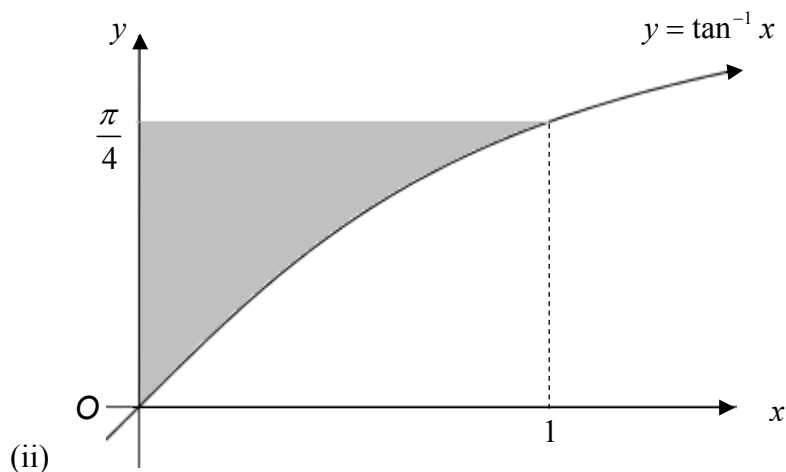
Start a new booklet.

Question 6 (12 marks).

Marks

- a) (i) Find $\frac{d}{dy}(\ln \cos y)$.

1



Show that the shaded area is given by $A = \frac{1}{2} \ln 2$ units²

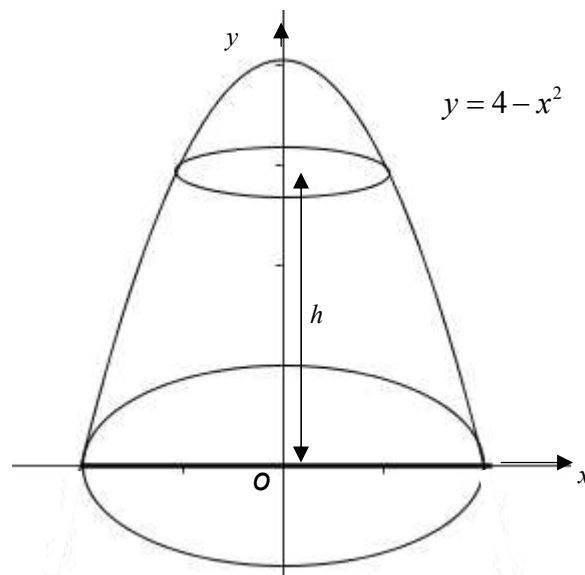
3

- b) P, Q, R and S are four points taken in order on a circle. Prove that:

$$\frac{PR}{QS} = \frac{\sin \hat{PQR}}{\sin \hat{QPS}}$$

3

Question 6 continued next page.

Question 6 continued**c)**

A mould for a container is made by rotating the part of the curve $y = 4 - x^2$ which lies in the first quadrant through one complete revolution about the y -axis. After sealing the base of the container, water is poured through a hole in the top. When the depth of water in the container is h cm, the depth is

changing at a rate of $\frac{10}{\pi(4-h)} \text{ cms}^{-1}$.

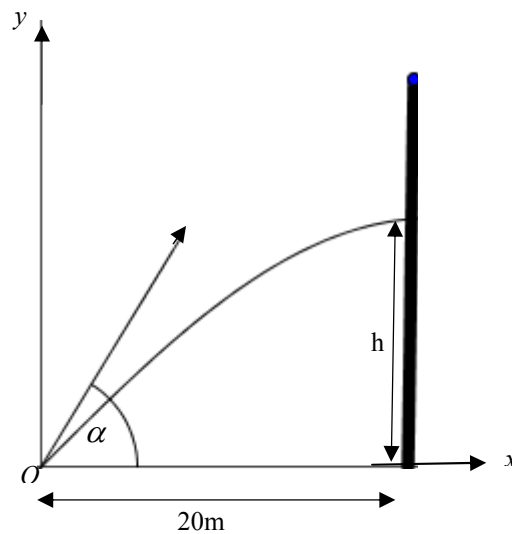
- (i) Show that when the depth is h cm, the surface area $S \text{ cm}^2$ of the top of the water is given by $S = \pi(4-h)$. 2
- (ii) Find the rate at which the surface area of the water is changing when the depth of the water is 2cm. 3

End of Question 6.

Start a new booklet.

Question 7 (12 marks).**Marks**

- a) A softball player hits the ball from ground level with a speed of 20 m/s and an angle of elevation α . It flies toward a high wall 20m away on level ground. Taking the origin at the point where the ball is hit, the derived expressions for the horizontal and vertical components of x and y of displacement at the time t seconds, taking $g = 10 \text{ m/s}^2$, are $x = 20t \cos \alpha$ and $y = -5t^2 + 20t \sin \alpha$



- (i) Hence find the equation of the path of the ball in flight in terms of x , y and α . 1
- (ii) Show that the height h at which the ball hits the wall is given by $h = 20 \tan \alpha - 5(1 + \tan^2 \alpha)$ 1
- (iii) Using part (ii) above, show that the maximum value of h occurs when $\tan \alpha = 2$ and find this maximum height 2

Question 7 continued next page.

Question 7 continued

- b)** A particle of unit mass moves in a straight line. It is placed at the origin on the x -axis and is then released from rest. When at position x , its acceleration is given by:

$$-9x + \frac{5}{(2-x)^2}.$$

Prove that the particle ultimately moves between two points on the x -axis and find these points.

3

- c)** (i) For any angles α and β show that

$$\tan \alpha + \tan \beta = \tan(\alpha + \beta)[1 - \tan \alpha \tan \beta]$$

1

- (ii) Prove, by mathematical induction, that

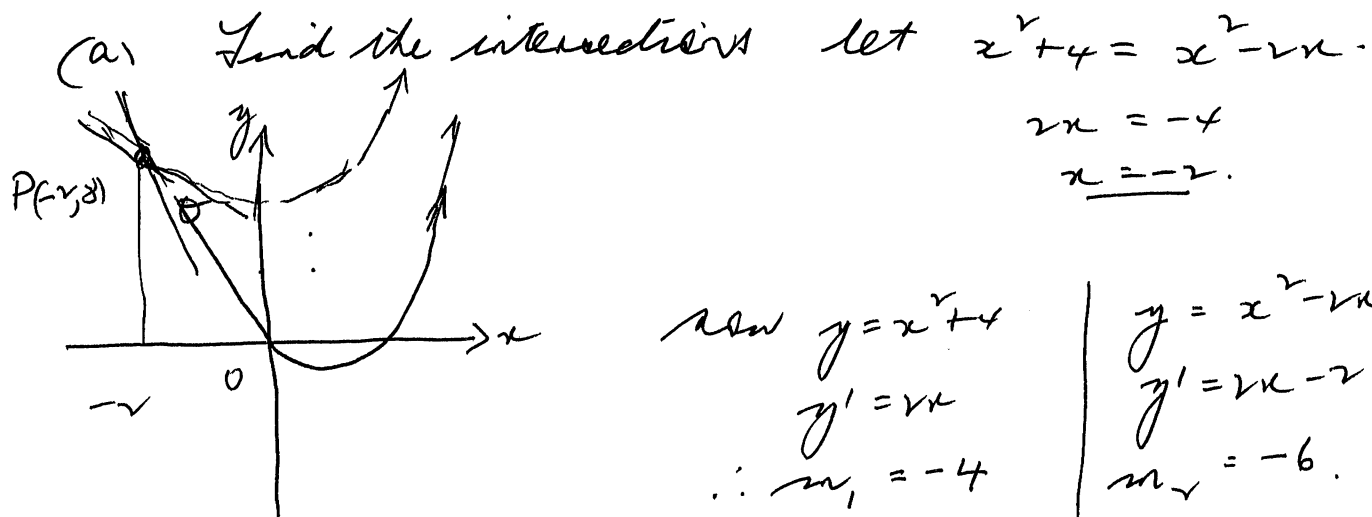
$$\tan \theta \tan 2\theta + \tan 2\theta \tan 3\theta + \dots + \tan n\theta \tan (n+1)\theta = \tan (n+1)\theta \cot \theta - (n+1)$$

for all positive integers n

4

End of Question 7.**End of Examination.**

QUESTION 1. (X1)



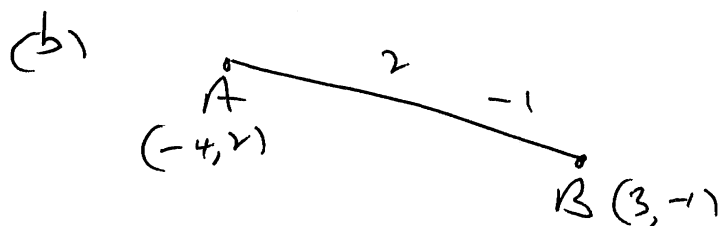
$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{-4 - (-6)}{1 + (-4)(-6)} \right|$$

$$= \left| \frac{-4 + 6}{1 + 24} \right|$$

$$= \frac{2}{25}$$

$$\therefore \theta = \tan^{-1} \frac{2}{25} = 4^\circ 34'$$



$$P = \left(\frac{2 \times 3 + (-1) \times (-4)}{2 + (-1)}, \frac{2 \times (-1) + (-1) \times 2}{2 + (-1)} \right)$$

$$= \boxed{(10, -4)}$$

(c) $y = \ln(\sin^2 x)$

$$y' = \frac{\frac{1}{\sqrt{1-x^2}}}{\sin^2 x}$$

$$= \boxed{\frac{1}{\sin^2 x \sqrt{1-x^2}}}$$

$$(d) \quad \frac{x-1}{x+3} \geq -2$$

$$\frac{x-1}{x+3} + 2 \geq 0$$

$$\frac{x-1+2(x+3)}{x+3} \geq 0$$

$$\frac{x-1+2x+6}{x+3} \geq 0$$

$$\frac{3x+5}{x+3} \geq 0$$

$$\frac{(3x+5)}{(x+3)} \times (x+3)^2 \geq 0$$

$$(3x+5)(x+3) \geq 0$$

$$\begin{array}{c} \leftarrow 0 \qquad \qquad 0 \rightarrow \\ -3 \qquad \qquad -\frac{5}{3} \end{array}$$

$$\therefore \boxed{x < -3, x \geq -\frac{5}{3}}$$

(e)

$$\cos 2B = \cancel{2} 1 - 2 \sin^2 B$$

$$= 1 - 2 \times \left(\frac{1}{3}\right)^2$$

$$= 1 - \frac{2}{9}$$

$$= \frac{7}{9}$$

$$\therefore \cos 2B = \frac{7}{9} = \cos A$$

$$\therefore \boxed{A = 2B}$$

[NB Doing this question on a calculator is not a fail]

$$(f) \quad u = t+1$$

$$du = dt$$

$$\therefore \int_0^1 \frac{t}{\sqrt{1+t}} dt = \int_{1/2}^2 \frac{u-1}{\sqrt{u}} du$$

$$= \int (u^{1/2} - u^{-1/2}) du$$

$$= \left[\frac{2}{3} u^{3/2} - 2 u^{1/2} \right]_{1/2}^2 = \frac{2}{3} \cdot 2^{3/2} - 2 \cdot 2^{1/2} - \left(\frac{2}{3} \cdot \left(\frac{1}{2}\right)^{3/2} - 2 \cdot \left(\frac{1}{2}\right)^{1/2} \right)$$

$$= \frac{4}{3} - \frac{2\sqrt{2}}{3}$$

$$\boxed{\frac{4-2\sqrt{2}}{3}}$$

Question 2

$$a) P(x) = ax^3 + bx^2 - 8x + 3$$

$$P(1) = 0$$

$$\therefore 0 = a + b - 8 + 3$$

$$\boxed{a + b = 5} \quad (1)$$

$$P(-2) = 15$$

$$15 = -8a + 4b^2 + 16 + 3$$

$$\boxed{8a - 4b = 4} \quad (2)$$

$$(1) \times 4$$

$$4a + 4b = 20 \quad (3)$$

$$(2) + (3)$$

$$12a = 24$$

$$\boxed{a = 2} \quad 1 \text{ mark}$$

sub into (1)

$$2 + b = 5$$

$$\boxed{b = 3} \quad 1 \text{ mark}$$

$$\therefore P(x) = 2x^3 + 3x^2 - 8x + 3$$

$$2x^2 + 5x - 3$$

$$x-1 \overline{) 2x^3 + 3x^2 - 8x + 3}$$

$$2x^3 - 2x^2$$

$$5x^2 - 8x + 3$$

$$5x^2 - 5x$$

$$-3x + 3$$

$$-3x + 3$$

$$\underline{\underline{0}}$$

$$2x^2 + 5x - 3$$

$$= \frac{(2x+6)(2x-1)}{2}$$

$$2$$

$$= \frac{2(x+3)(2x-1)}{2}$$

$$\neq$$

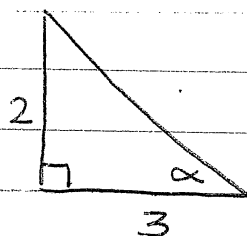
$$\therefore P(x) = (x-1)(x+3)(2x-1)$$

1 mark

$$b) i) 3\sin\theta + 2\cos\theta = R\sin(\theta + \alpha)$$

$$R = \sqrt{3^2 + 2^2}$$

$$= \sqrt{13} \quad 1 \text{ mark}$$



$$\tan\alpha = \frac{2}{3}$$

$$\alpha = \tan^{-1} 2/3 = 33^\circ 41'$$

1 mark

$$ii) \sqrt{13} \sin(\theta + 33^\circ 41') = \frac{5}{2}$$

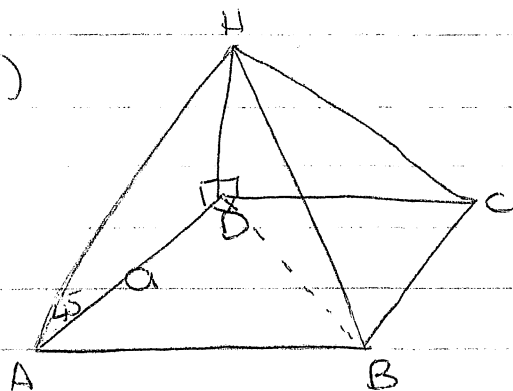
$$\sin(\theta + 33^\circ 41') = \frac{5}{2\sqrt{13}}$$

$$\theta + 33^\circ 41' = \sin^{-1} \frac{5}{2\sqrt{13}}$$

$$\theta = \sin^{-1} \frac{5}{2\sqrt{13}} - 33^\circ 41'$$

$$\theta = 10^\circ 13', 102^\circ 25' \quad 1 \text{ mark each}$$

c)



$$\tan \Theta = a/2a$$

$$\tan \Theta = 1/2$$

$$\Theta = 26^{\circ}33'54.18''$$

$$= 26^{\circ}34' \text{ (nearest min)}$$

1 mark

$$d) a_1 = a - \frac{f(a)}{f'(a)}$$

$$f(x) = 2x + \cos x$$

$$f'(x) = 2 - \sin x$$

$$a = -\pi/6$$

$$f(a) = \frac{-\pi}{3} + \frac{\sqrt{3}}{2}$$

$$f'(a) = 2 - -1/2$$

$$= 5/2$$

$$a_1 = -\pi/6 - \left[\frac{-2\pi + 3\sqrt{3}}{6} \right] \div \frac{5}{2}$$

1 mark

$$= \frac{-\pi}{6} - \left[\frac{-4\pi + 6\sqrt{3}}{30} \right]$$

$$= \frac{-5\pi}{30} + \frac{4\pi - 6\sqrt{3}}{30}$$

$$= \frac{-\pi - 6\sqrt{3}}{30} \text{ 1 mark}$$

$$i) \angle AHD = 45^{\circ}$$

$\therefore \Delta AHD$ is isosceles

$$\therefore HD = a$$

In Δ 's HCD + DBA

$$HD = a = DA$$

$$\angle HDC = 90^{\circ} = \angle DAB$$

(given & properties of a rectangle)

$$DC = AB \text{ (opposite sides}$$

of a rectangle)

$$\therefore \Delta HCD \equiv \Delta DBA \text{ (SAS)}$$

$$\therefore \angle DBA = 30^{\circ}$$

1 mark

In ΔOAB

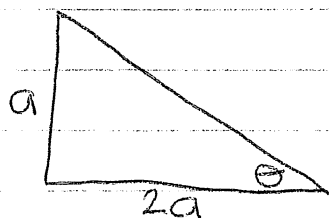
$$\sin 30 = \frac{a}{BD}$$

$$1/2 = a/BD$$

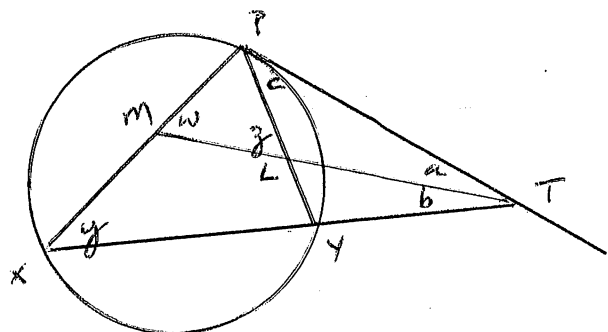
$$\therefore BD = 2a$$

1 mark

$$ii) \Delta HDB$$



Question 3:



$a = b$ (TM bisects $\angle PTX$, given)
 $c = y$ (alternate segment theorem)
 $z = a + c$ (exterior $\angle$ of $\triangle PLY$)
 $= b + y$
 $w = b + y$ (exterior $\angle$ of $\triangle MTX$)
 $\therefore z = w$
 $\therefore \triangle PLM$ is isosceles (base angles equal)

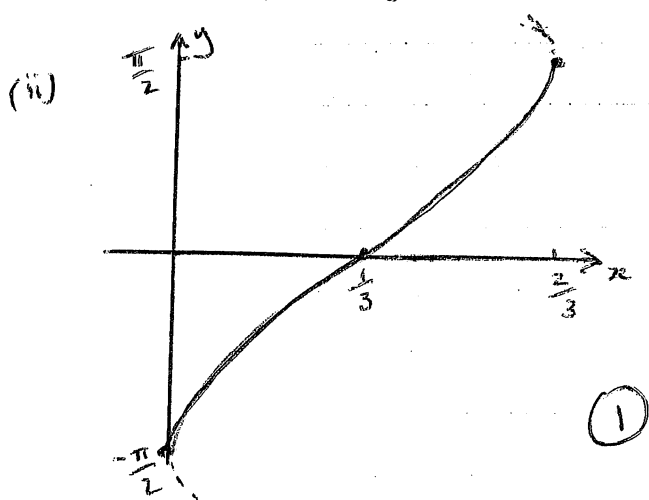
(3)

(b) (i) $f^{-1}(x) = \sin^{-1}(3x-1)$

Domain: $-1 \leq 3x-1 \leq 1$
 $0 \leq 3x \leq 2$
 $0 \leq x \leq \frac{2}{3}$

(2)

Range: $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$



(1)

(iii) $x = \sin^{-1}(3y-1)$
 $\sin x = 3y-1$
 $3y = \sin x + 1$
 $y = \frac{1}{3}(\sin x + 1)$
 $f(x) = \frac{1}{3}(\sin x + 1)$

Domain: $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$

Range: $0 \leq y \leq \frac{2}{3}$

(3)

(c) (i) $\frac{dT}{dt} = \frac{d}{dt}(T_0 + Ae^{-kt})$
 $= Ae^{-kt} \cdot (-k)$
 $= -(T - T_0) \cdot (-k)$
 $= -k(T - T_0)$

(1)

(ii) When $t=0$: $T = 85$

$\therefore 85 = 25 + A$

$\therefore A = 60$

$\therefore T = 25 + 60e^{-kt}$

When $t=1$: $80 = 25 + 60e^{-k}$

$\therefore 55 = 60e^{-k}$

$\therefore e^{-k} = \frac{55}{60}$

$k = -\ln\left(\frac{55}{60}\right)$

When $t=5$: $T = 25 + 60e^{-5k}$

$= 63.8336 \dots$

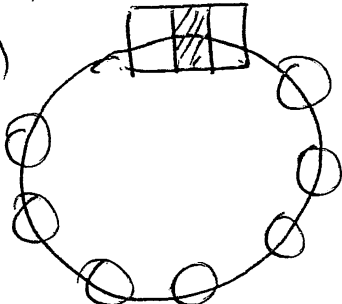
$\approx 64^\circ$

(2)



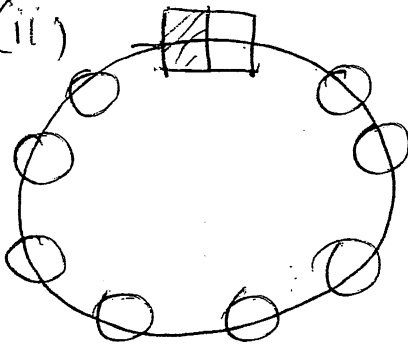
Solution to Q(4)

(a)

(i)  1

$7! \times 1 \times 2!$
 $\uparrow \quad \quad \uparrow \quad \quad \uparrow$
 the rest particular girl 2 boys

$= 2 \times 7!$
 $= 10080$

(ii) 

Sitting together
 $= 8! \times 2!$
 $\therefore \text{Sedanto} = 91 - 8 \times 2!$

$$= 282240 \quad [1]$$

(b) $y = x^2/4a$

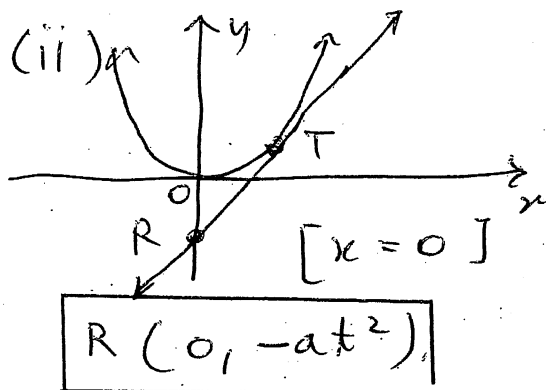
(i) $\frac{dy}{dx} = \frac{x}{2a}$ 2

$\left. \frac{dy}{dx} \right|_{x=2at} = \frac{2at}{2a} = t$

$\therefore$ Equ. of tangent

$y - at^2 = t(x - 2at)$

$y = tx - at^3$ 1



If PQ passes through R, then
Coordinates of R
 satisfy equation of PQ

i.e. $-at^2 + apq = 0$ ✓

$t^2 = pq$

i.e. $\frac{t}{p} = \frac{q}{t}$ ✓ 2

$\therefore p, t, q$ are terms of a geometric series

(c) $n = 2 \cos 3t, \dot{x} = -6 \sin 3t$

$\dot{x} = -9(2 \cos 3t) = -9x$

(i) $\therefore n^2 = 9$

(ii) $T = \frac{2\pi}{n} = \frac{2\pi}{3}$ 1

(iii) $\omega \rightarrow 3t = \frac{1}{2} \therefore 3t = \frac{\pi}{3}, \frac{5\pi}{3}$

$\Rightarrow t = \frac{\pi}{9}, \frac{5\pi}{9}$

$\therefore \dot{x} = -6 \sin \frac{\pi}{3}, 6 \sin \frac{5\pi}{3}$

$= \pm 3\sqrt{3} \text{ cm/sec}$

$\approx \pm 5.196 \text{ cm/sec}$ 2

QUESTION 5.

$$(a) \frac{\sin \theta}{\cos \theta} = 2 \sin \theta \cos \theta$$

$$\sin \theta = 2 \sin \theta \cos^2 \theta.$$

$$2 \sin \theta (1 - \sin^2 \theta) - \sin \theta = 0.$$

$$2 \sin^3 \theta - \sin \theta = 0.$$

$$\sin \theta (2 \sin^2 \theta - 1) = 0.$$

$$\sin \theta = 0$$

$$\theta = \pi n \quad n \in \mathbb{Z}$$

$$\sin \theta = \pm \frac{1}{\sqrt{2}}$$

$$\theta = \pi n \pm \frac{\pi}{4} \quad n \in \mathbb{Z}$$

$$(b) (i) \alpha \beta + \beta \gamma + \alpha \gamma = \frac{1}{2}.$$

$$(ii) \alpha \beta \gamma = \frac{1}{2}.$$

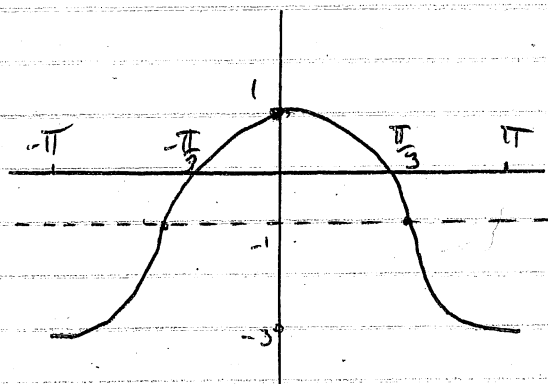
$$\frac{1}{\cos a} + \frac{1}{\cos b} + \frac{1}{\cos c} =$$

$$\frac{\cos a \cos b + \cos a \cos c + \cos b \cos c}{\cos a \cos b \cos c}.$$

$$= \frac{\frac{1}{2}}{\frac{1}{2}}$$

$$= 1$$

(c) (i)



$$2 \cos x - 1 = 0$$

$$\cos x = \frac{1}{2}$$

$$x = \pm \frac{\pi}{3}.$$

(ii)

$$V = \pi \int_{-\pi/3}^{\pi/3} 4\cos^2 x - 4\cos x + 1 \, dx.$$

$$= \pi \int_{-\pi/3}^{\pi/3} 2\cos 2x + 2 - 4\cos x + 1 \, dx.$$

$$= \pi \left[\sin 2x - 4\sin x + 3x \right]_{-\pi/3}^{\pi/3}.$$

$$= 2\pi \left(\sin \frac{2\pi}{3} - 4\sin \frac{\pi}{3} + \pi \right).$$

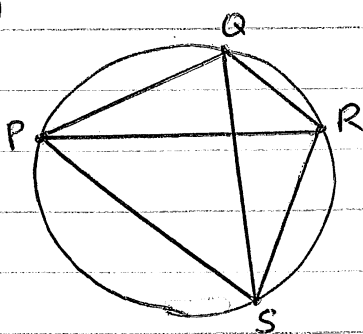
$$= 2\pi^2 - 3\pi\sqrt{3} \text{ units}^3.$$

Question 6

$$\begin{aligned} \text{a) i) } \frac{d(\ln \cos y)}{dy} &= \frac{-\sin y}{\cos y} \\ &= \underline{\underline{-\tan y}} \end{aligned}$$

$$\begin{aligned} \text{ii) } A &= \int_0^{\frac{\pi}{4}} x \, dy \\ &= \int_0^{\frac{\pi}{4}} \tan y \, dy \\ &= - \int_0^{\frac{\pi}{4}} -\tan y \, dy \\ &= - \left[\ln(\cos y) \right]_0^{\frac{\pi}{4}} \quad \text{using (i)} \\ &= - \left(\ln \left(\cos \frac{\pi}{4} \right) - \ln(\cos 0) \right) \\ &= - \ln \left(\frac{1}{\sqrt{2}} \right) + \ln 1 \\ &= - \ln(2)^{-\frac{1}{2}} \\ &= \frac{1}{2} \ln 2 \quad \text{units}^2 \end{aligned}$$

b)



In $\triangle PQR$

$$\frac{\sin \hat{PQR}}{PR} = \frac{\sin \hat{QRP}}{PQ}$$

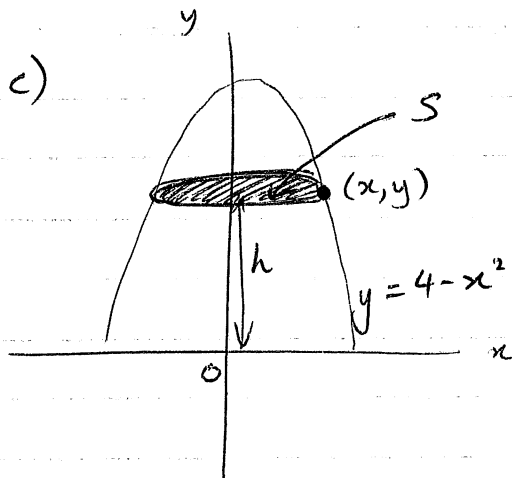
In $\triangle QPS$

$$\frac{\sin \hat{QPS}}{QS} = \frac{\sin \hat{PSQ}}{PQ}$$

$\hat{QRP} = \hat{PSQ}$ (angles in same segment (arc PQ))

$$\therefore \frac{\sin \hat{PQR}}{PR} = \frac{\sin \hat{QPS}}{QS}$$

$$\frac{\sin \hat{PQR}}{\sin \hat{QPS}} = \frac{PR}{QS}$$



i) $S = \pi r^2$
 $S = \pi x^2$

when $y = h$

$$h = 4 - x^2$$

$$x^2 = 4 - h$$

$$\therefore S = \pi(4 - h)$$

ii) $S = 4\pi - \pi h$

$$\frac{dS}{dh} = -\pi$$

$$\frac{dS}{dt} = \frac{dS}{dh} \times \frac{dh}{dt}$$

$$= -\pi \times \frac{10}{\pi(4-h)}$$

$$= -\frac{10}{4-h}$$

when $h = 2$

$$\frac{dS}{dt} = -\frac{10}{(4-2)}$$

$$= -5 \text{ cm}^2/\text{s}$$

QUESTION 7

(a) $x = 20t \cos \alpha$
 $y = -5t^2 + 20t \sin \alpha$

(i) $\Rightarrow y = -5 \left(\frac{x}{20 \cos \alpha} \right)^2 + 20 \left(\frac{x}{20 \cos \alpha} \right) \sin \alpha$

$y = -\frac{1}{80} x^2 \sec^2 \alpha + x \tan \alpha$

ie $y = -\frac{1}{80} (\tan^2 \alpha + 1) x^2 + (\tan \alpha) x$

(ii) When $x = 20$, $y = h$

$\Rightarrow h = -\frac{1}{80} (\tan^2 \alpha + 1) 400 + 20 \tan \alpha$

ie $h = -5 \tan^2 \alpha + 20 \tan \alpha - 5$

$h = 20 \tan \alpha - 5(1 + \tan^2 \alpha)$

(iii)

$h = -5 \tan^2 \alpha + 20 \tan \alpha - 5$

Max. value of h occurs

when $\tan \alpha = \frac{-20}{2(-5)} = 2$

ie $\tan \alpha = 2$

Max height is

$-5(2)^2 + 20(2) - 5 = 15 \text{ metres}$

(b) $\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = -9x + 5(x-2)^{-2}$

$\frac{1}{2} v^2 = -\frac{9x^2}{2} + \frac{5}{2-x} + C$

$\left. \begin{matrix} x=0 \\ v=0 \end{matrix} \right\} \Rightarrow C = -\frac{5}{2}$

$\therefore v^2 = -9x^2 + \frac{5}{2-x} - 5$

(b) $v^2 = -9x^2 + \frac{10}{2-x} - 5$

For motion to exist then

$v^2 \geq 0$

ie $-9x^2 + \frac{10}{2-x} - 5 \geq 0$

$-9x^2(2-x)^2 + 10(2-x) - 5(2-x)^2 \geq 0$

$(2-x) [-9x^2(2-x) + 10 - 5(2-x)] \geq 0$

ie $(2-x)(-18x^2 + 9x^3 + 5x) \geq 0$

ie $(2-x) \cdot x(9x^2 - 18x + 5) \geq 0$

$x(2-x)(3x-5)(3x-1) \geq 0$

SOLUTION

$0 \leq x \leq \frac{1}{3}$ $\frac{5}{3} \leq x \leq 2$

However since particle starts at zero and changes direction at $x = \frac{1}{3}$ it can never be outside the interval $0 \leq x \leq \frac{1}{3}$.

Note For $\frac{1}{3} < x < \frac{5}{3}$ $v^2 < 0$

* impossible to move in this interval and therefore cannot move in $\frac{5}{3} \leq x \leq 2$.

Ultimately moves in interval

$0 \leq x \leq \frac{1}{3}$

⑦ (c) RHS = $\tan(\alpha + \beta) [1 - \tan \alpha \tan \beta]$
 (i) $= \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} [1 - \tan \alpha \tan \beta]$
 $= \tan \alpha + \tan \beta$
 $= \text{LHS}$

(ii) When $n=1$ $\tan \theta \cdot \tan 2\theta = \tan 2\theta \cot \theta - 2$
 $\text{RHS} = \frac{2 \tan \theta}{1 - \tan^2 \theta} \cdot \frac{1}{\tan \theta} - 2 = \frac{2 \tan^2 \theta}{1 - \tan^2 \theta} = \tan \theta \cdot \frac{2 \tan \theta}{1 - \tan^2 \theta}$
 $= \tan \theta \cdot \tan 2\theta = \text{LHS}$

- Assume $\tan \theta \tan 2\theta + \dots + \tan k\theta \tan (k+1)\theta = \tan (k+1)\theta \cot \theta - (k+1)$

RTP $\tan \theta \tan 2\theta + \dots + \tan k\theta + \tan (k+1)\theta + \tan (k+1)\theta \tan (k+2)\theta = \tan (k+2)\theta \cot \theta - (k+2)$

Now LHS = $\tan (k+1)\theta \cdot \cot \theta - (k+1) + \tan (k+1)\theta \tan (k+2)\theta$
 $= \cot \theta [\tan (k+1)\theta + \tan (k+1)\theta \cdot \tan (k+2)\theta \cdot \tan \theta] - (k+1)$
 $= \cot \theta [\tan (k+1)\theta + \tan (k+2)\theta \left(1 - \frac{\tan (k+1)\theta + \tan \theta}{\tan (k+2)\theta}\right)] - (k+1)$
 $= \cot \theta [\tan (k+1)\theta + \tan (k+2)\theta - \tan (k+1)\theta - \tan \theta] - (k+1) \quad \leftarrow \text{using (i)}$
 $= \cot \theta [\tan (k+2)\theta - \tan \theta] - (k+1)$
 $= \cot \theta [\tan (k+2)\theta] - 1 - (k+1)$
 $= \cot \theta [\tan (k+2)\theta] - (k+2) = \text{RHS}$